

Smart Water Tank Management System

- LPC2129 ARM7
- ADC & Ultrasonic
- I2C RTC / EEPROM
- ESP-01 ThingSpeak
- CAN Protocol

1. Project Overview

The **Smart Water Tank Management System** is an industrial-grade embedded monitoring and control solution based on the NXP **LPC2129 ARM7** microcontroller. The architecture integrates multi-sensor telemetry (water level, temperature, and ultrasonic distance), real-time timestamping via an I2C RTC, non-volatile data logging via I2C EEPROM, automated/manual relay pump actuation, local visual display on a 16x2 LCD, remote IoT telemetry via ESP-01 Wi-Fi (ThingSpeak cloud integration), and high-reliability local bus communication via the CAN protocol.

2. Functional Architecture

The LPC2129 acts as the master central controller, orchestrating sensor data acquisition, relay safety logic, local user feedback, data persistence, and dual-network communications.

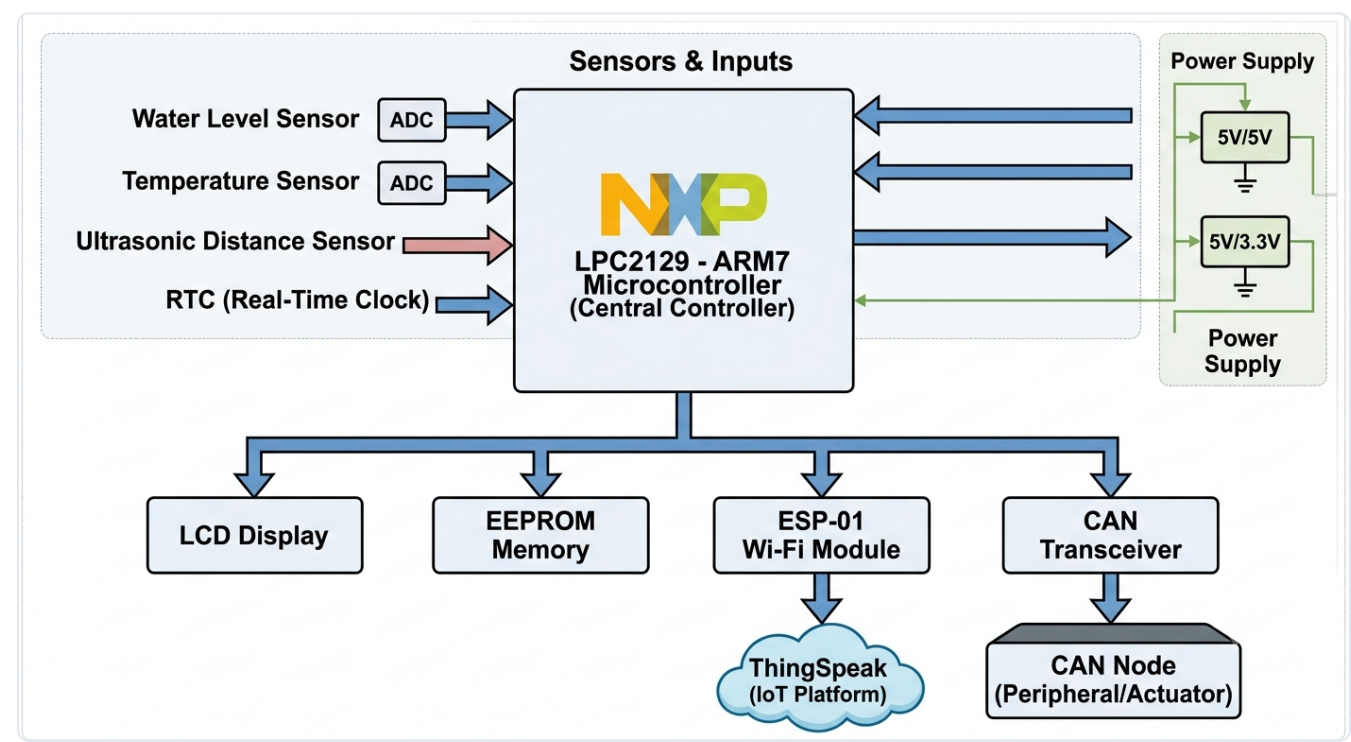


Figure 1: Hardware System Architecture & Pin Interconnect Block Diagram

3. System Initialization Sequence

Upon system boot, the firmware initializes core peripherals prior to entering the main execution loop:

- **UART0 Diagnostics:** Configured for debug logging at standard baud rate.
- **UART1 / ESP-01:** Baud rate calculated from VPBDIV and initialized to 115,200 bps.
- **Ultrasonic Sensor Pins:** P0.12 configured as GPIO Output (TRIG) and P0.13 as GPIO Input (ECHO).
- **I2C & ADC Peripherals:** Master I2C initialized for RTC & EEPROM communication; ADC channels enabled.
- **LCD & External Interrupts:** 16x2 LCD bus initialized and Pushbutton Interrupts mapped for manual control.

4. Main Control Loop

The primary operational pipeline operates continuously to perform the following sequence:

1. Poll for manual mode triggers (Pushbuttons SW1 and SW2).
2. Sample analog inputs: Read Water Level (ADC Ch 0) and Temperature (ADC Ch 2).
3. Check RTC half-hour log boundary condition (00 or 30 min/sec).
4. Render current status on the 16x2 LCD screen and forward diagnostic text to UART0.
5. Maintain cloud connectivity via ESP-01 and transmit updated field parameters to ThingSpeak.
6. Perform a scheduled 20-second delay cycle prior to executing EEPROM readbacks and issuing CAN frame transmissions.

5. Actuator & Pump Control Logic

Relay drive outputs are mapped to pins `P0.10` and `P0.21` via the function `smart_tank_relay()` :

- **Pump ON State (Arg 1):** Asserts both `P0.10` and `P0.21` HIGH. Records status `0x01` at EEPROM address `0x0B` .
- **Pump OFF State (Arg 0):** Clears both outputs LOW. Records status `0x00` at EEPROM address `0x0B` .

Automatic Threshold Conditions:

- *Low Water Level (Turn Relay ON):* `ADC < 200`
- *Tank Full Level (Turn Relay OFF):* `ADC ≥ 800`

6. Water-Level Sensing (ADC Channel 0)

Water level measurement uses ADC Channel 0. Raw 10-bit conversion values are evaluated directly against threshold limits for control decisions, and percentage water level is normalized using:

$$\text{Level (\%)} = [(ADC_Raw - 120) \times 100] / (900 - 120)$$

7. Ultrasonic Distance Measurement

Distance sensing is performed using high-speed GPIO pulse timing. The sequence starts with a precise trigger pulse and counts the pulse width returned on the ECHO pin.

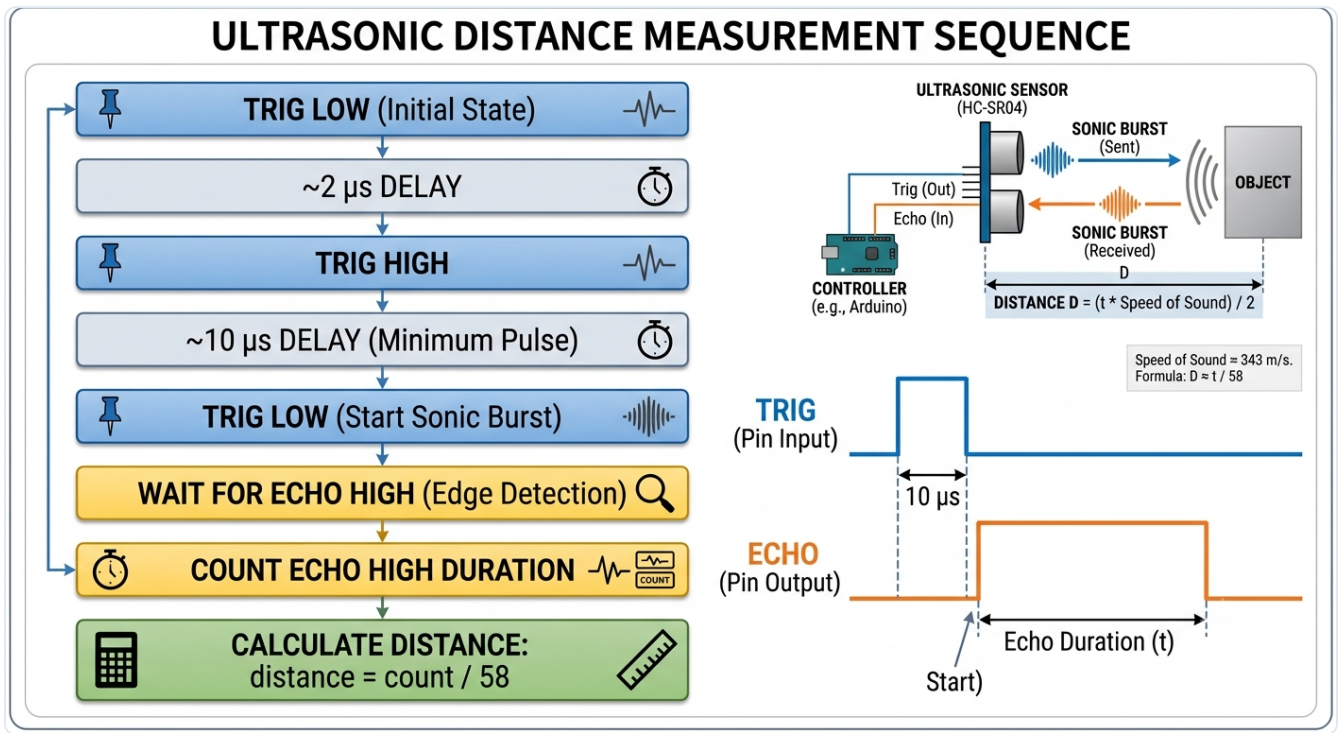


Figure 2: Ultrasonic Trigger Timing & Distance Pulse Measurement Flow

Distance Formula: $\text{Distance (cm)} = \text{Echo_Count} / 58$

8. Temperature Sensing (ADC Channel 2)

Ambient/Water temperature is sampled via ADC Channel 2 utilizing standard linear analog transducer scaling:

$$V_{\text{out}} = (\text{ADC_Raw} \times 3.3\text{V}) / 1023$$

$$\text{Temperature } (^{\circ}\text{C}) = (V_{\text{out}} - 0.5\text{V}) / 0.01\text{V}/^{\circ}\text{C}$$

9. Real-Time Clock (RTC) Integration

An external RTC is interfaced via the I2C bus. The microcontroller retrieves BCD-encoded values for Seconds, Minutes, Hours, Date, Month, and Year. BCD values are converted to standard integers for display and saved to non-volatile EEPROM storage. AM/PM designator text is maintained at address `0x03`.

10. Data Logging Memory Map (EEPROM Address 0xA0)

The system utilizes an I2C EEPROM at slave address `0xA0` for persistent parameter logging.

EEPROM Address	Stored Parameter Description	Data Type / Size
0x00	RTC Seconds	8-bit Integer (1 Byte)
0x01	RTC Minutes	8-bit Integer (1 Byte)
0x02	RTC Hours	8-bit Integer (1 Byte)
0x03	AM / PM Designator	ASCII Character (1 Byte)
0x04	Date	8-bit Integer (1 Byte)
0x05	Month	8-bit Integer (1 Byte)
0x06	Year	8-bit Integer (1 Byte)
0x07 - 0x08	Water Level ADC (Low Byte / High Byte)	16-bit Integer (2 Bytes)
0x09	Ultrasonic Distance	8-bit Integer (1 Byte)
0x0A	Temperature (°C)	8-bit Integer (1 Byte)
0x0B	Pump Actuation Status (1 = ON, 0 = OFF)	8-bit Integer (1 Byte)

11. CAN Bus Protocol & Frame Formats

Local device-to-device communication utilizes Controller Area Network (CAN) messaging across two distinct 8-byte frame structures:

CAN ID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
0x302 (Time/Date)	Hour	Minute	Second	AM/ PM	Date	Month	Year	Reserved
0x301 (Tank Telemetry)	Water (L)	Water (H)	Distance	Temp	Pump Status	Reserved	Reserved	Reserved

12. Wi-Fi Connectivity & Cloud Telemetry (ThingSpeak)

Remote IoT integration is facilitated via an ESP-01 Wi-Fi module over UART1 (115,200 baud). The module establishes a TCP connection to api.thingspeak.com:80 and sends telemetry data using standard HTTP GET requests.

ThingSpeak Field	Mapped Parameter Source
Field 1	Water Level Percentage / Raw ADC Value
Field 2	Ultrasonic Distance Measurement (cm)
Field 3	Temperature Sensor Value (°C)
Field 4	Timestamp Format (HHMMSS)
Field 5	Datestamp Format (DDMMYY)
Field 6	Pump State Identifier (0 = OFF, 1 = ON)

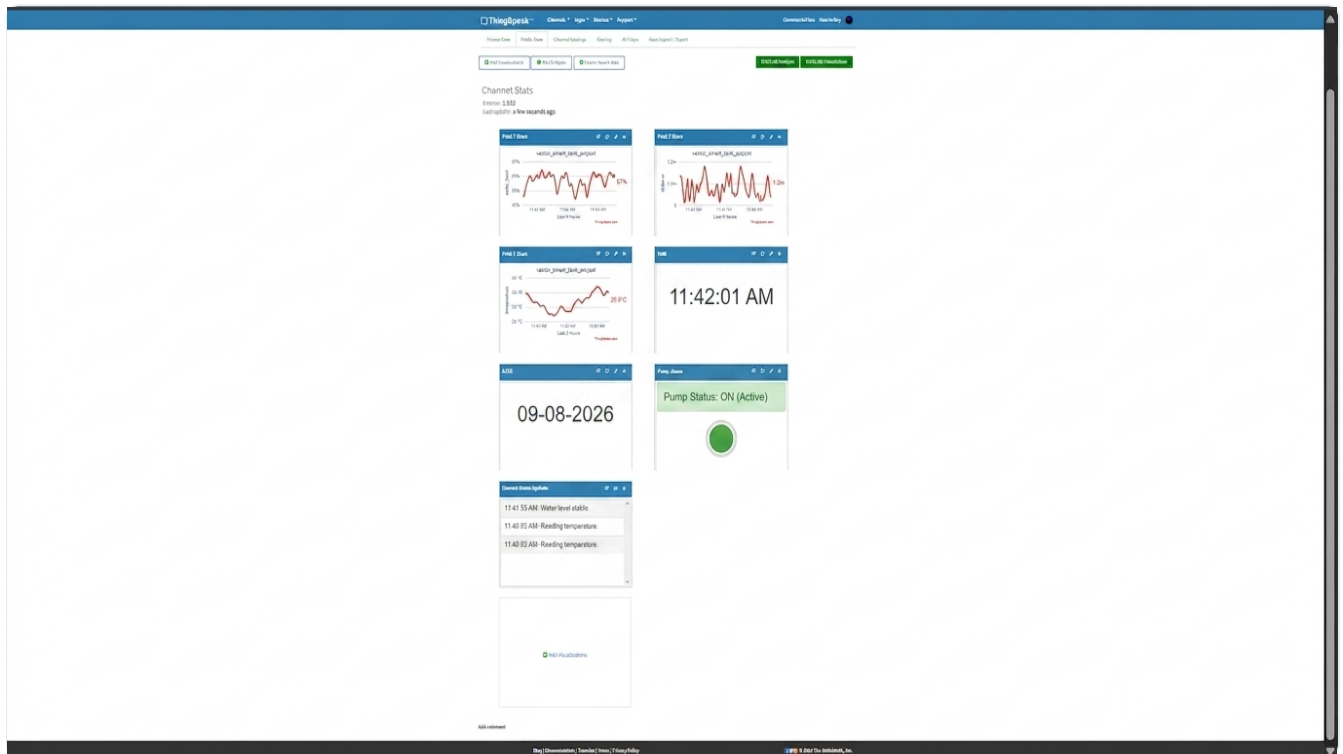


Figure 3: ThingSpeak Live IoT Telemetry Channel Stats Dashboard (channel/3444987)

13. Peripheral Hardware Mapping Summary

Peripheral Function	Interface / LPC2129 Pins	Purpose / Description
Water Level Sensor	ADC Channel 0	Analog water depth monitoring
Temperature Sensor	ADC Channel 2	Water/Ambient temperature sampling
Ultrasonic Transducer	GPIO P0.12 (TRIG), P0.13 (ECHO)	Non-contact distance sensing
Manual Pushbutton SW1	GPIO P0.14	Manual override: Pump ON
Manual Pushbutton SW2	GPIO P0.15	Manual override: Pump OFF / Exit
ESP-01 Wi-Fi Module	UART1 (P0.8 TXD1, P0.9 RXD1)	Cloud connectivity & ThingSpeak updates
RTC & EEPROM	I2C Bus (SCL / SDA)	Real-time clock & non-volatile logging
Pump Actuator / Relays	GPIO P0.10, P0.21	Dual-channel pump switching outputs
CAN Bus Interface	CAN1 Controller	Industrial binary network telemetry